

Poisson regression – three views

1. Index form

The per-observation reading: what happens for each row i of your data.

$$\begin{aligned} y_i \mid \mu_i &\sim \text{Poisson}(\mu_i) \\ \log(\mu_i) &= \beta_0 + \beta_1 x_i \end{aligned}$$

2. Matrix form

The same model in matrix notation. The structural contract every textbook past chapter 4 switches to.

$$\begin{aligned} \mathbf{y} \mid \boldsymbol{\mu} &\sim \text{Poisson}(\boldsymbol{\mu}) \\ \log(\boldsymbol{\mu}) &= \mathbf{X}\boldsymbol{\beta} \end{aligned}$$

3. Worked observation ($i = 1$)

The index-form equations evaluated at the first observation of your data, with coefficient estimates plugged in.

$$\underbrace{y_1}_{\text{observed}=1} = 0.955 + -0.0438 \cdot 0.45 + 0.0643 = 1$$

Predicted $\hat{\mu}_1 = 0.936$; residual $\hat{\varepsilon}_1 = 0.0643$.

Stacking the same response equation for all $n = 100$ observations:

$$\underbrace{\begin{bmatrix} 1 \\ 1 \\ 2 \\ 3 \\ 1 \\ \vdots \\ 0 \\ 5 \end{bmatrix}}_{\mathbf{y}_{100 \times 1} \text{ (observed)}} = \underbrace{\begin{bmatrix} 1 & 0.45 \\ 1 & -0.0186 \\ 1 & -0.318 \\ 1 & -0.929 \\ 1 & -1.49 \\ \vdots & \vdots \\ 1 & -0.166 \\ 1 & 1.02 \end{bmatrix}}_{\mathbf{X}_{100 \times 2}} \underbrace{\begin{bmatrix} 0.955 \\ -0.0438 \end{bmatrix}}_{\hat{\boldsymbol{\beta}}_{2 \times 1} \text{ (estimated)}} + \underbrace{\begin{bmatrix} 0.0643 \\ 0.0437 \\ 1.03 \\ 2 \\ -0.0207 \\ \vdots \\ -0.963 \\ 4.09 \end{bmatrix}}_{\hat{\boldsymbol{\varepsilon}}_{100 \times 1} \text{ (residual)}}$$